

networks, in particular wireless sensor networks. The tiny sensors of WSNs suffer from the lack of processing, memory and battery power. Applying any encryption scheme requires transmission of extra bits, hence extra processing, memory and battery power, which are very important resources for the sensors' longevity. Applying the security mechanisms such as encryption could also increase delay, jitter and packet loss in wireless sensor networks.

Steganography. While cryptography aims at hiding the content of a message, steganography hides the existence of the message. Steganography is the art of covert communication by embedding a message into the multimedia data (image, sound, video, etc). Its main objective is to modify the carrier in a way that is not perceptible and hence it looks just like ordinary. It is very useful when you want to send a secret data without sender information or you want to distribute secret data publicly.

Attacks in wireless sensor networks

A large-scale sensor network consists of thousands of sensor nodes and may be dispersed over a wide area. Typically, sensor nodes are susceptible to many kinds of attacks:

Passive information gathering. An adversary with powerful resources can collect information from the sensor networks if it is not encrypted.

Node subversion. Capture of a node may reveal its information including cryptographic keys and thus compromise the whole sensor network.

False node. Addition of a node by an adversary to inject malicious data, whereby the false node is computationally robust enough to lure other nodes to send data to it.

Node malfunction. A malfunctioning node generates inaccurate data which could jeopardise the integrity of the sensor network, especially if it is a data aggregating node such as a cluster leader.

Node outage. A node stops its function. In case a cluster leader stops func-

tioning, the sensor network protocols should be robust enough to mitigate the effects of node outages by providing an alternative route.

Message corruption. Any modification of the content of a message by an attacker compromises its integrity.

Traffic analysis. Even when the messages transferred are encrypted, it still leaves a high possibility of analysis of the communication patterns. Sensor activities can potentially reveal enough information to enable an adversary to cause malicious harm to the sensor network.

Sybil attack. In a Sybil attack, a single node presents multiple identities to other nodes in the network. They pose a significant threat to geographic routing protocols, where location-aware routing requires nodes to exchange coordinate information with their neighbours to efficiently route geographically addressed packets. Authentication and encryption techniques can prevent an outsider to launch a Sybil attack on the sensor network. Using globally shared keys allows an insider to masquerade as any (possibly even non-existent) node.

Sinkhole attack. In a sinkhole attack, the adversary's goal is to lure nearly all the traffic from a particular area through a compromised node, creating a metaphorical sinkhole with the adversary at the centre. Sinkhole attacks typically work by making a compromised node look especially attractive to surrounding nodes with respect to the routing algorithm.

Wormholes. In the wormhole attack, an adversary tunnels messages received in one part of the network over a low-latency link and replays them in a different part. The simplest instance of this attack is a single node situated between two other nodes forwarding messages between the two of them. However, wormhole attacks more commonly involve two distant malicious nodes colluding to understate their distance from each other by relaying packets along an out-of-bound channel available only to the attacker.

Key management

Key management is the process by which cryptographic keys are generated, stored, protected, transferred, loaded, used and destroyed. To achieve security in wireless sensor networks, it is important to be able to perform various cryptographic operations, including encryption, authentication and so on. Key management schemes are mechanisms used to establish and distribute various kinds of cryptographic keys in the network, such as individual keys, pairwise keys and group keys.

Key management is an essential cryptographic primitive upon which other security primitives are built. Most security requirements, such as privacy, authenticity and integrity, can be addressed by building a solid key management framework.

The challenge of designing key management protocols for sensor networks lies in establishing a secure communication infrastructure, before any routing fabric has been established with or without the presence of any trusted authority or fixed server, from a collection of sensor nodes that have no prior contact with each other. Some cryptographic information, e.g., a key, is normally preloaded in sensor nodes before deployment, and allows sensor nodes to perform secure communications with each other.

Trusted server schemes. Trusted server schemes depend on a trusted and secure server such as the base station for key agreement among nodes. The server can be treated as the key distribution centre (KDC). For example, assume that two sensor nodes intend to make a secure connection. In a typical case, a symmetric key is generated for each node in a sensor network before deployment and embedded in each sensor node's memory. This embedded key is used for the two sensors to authenticate themselves to the base station. Then the base station generates a link key or session key and sends it securely to both sensor nodes via a single hop or multiple hops. In the trusted server scheme the base station is the most appropriate